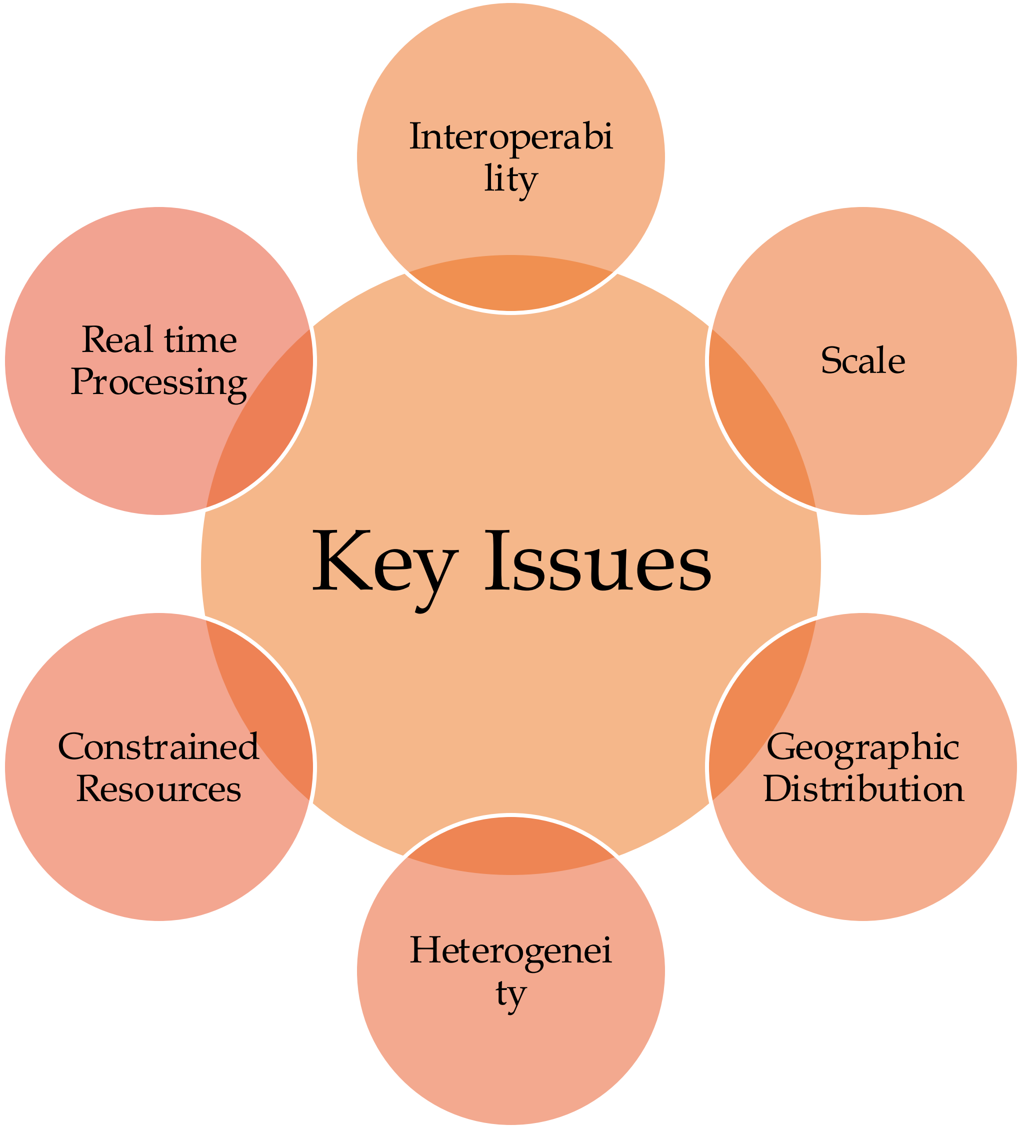
Fundamentals of Fog and Edge Computing

BCSE313L

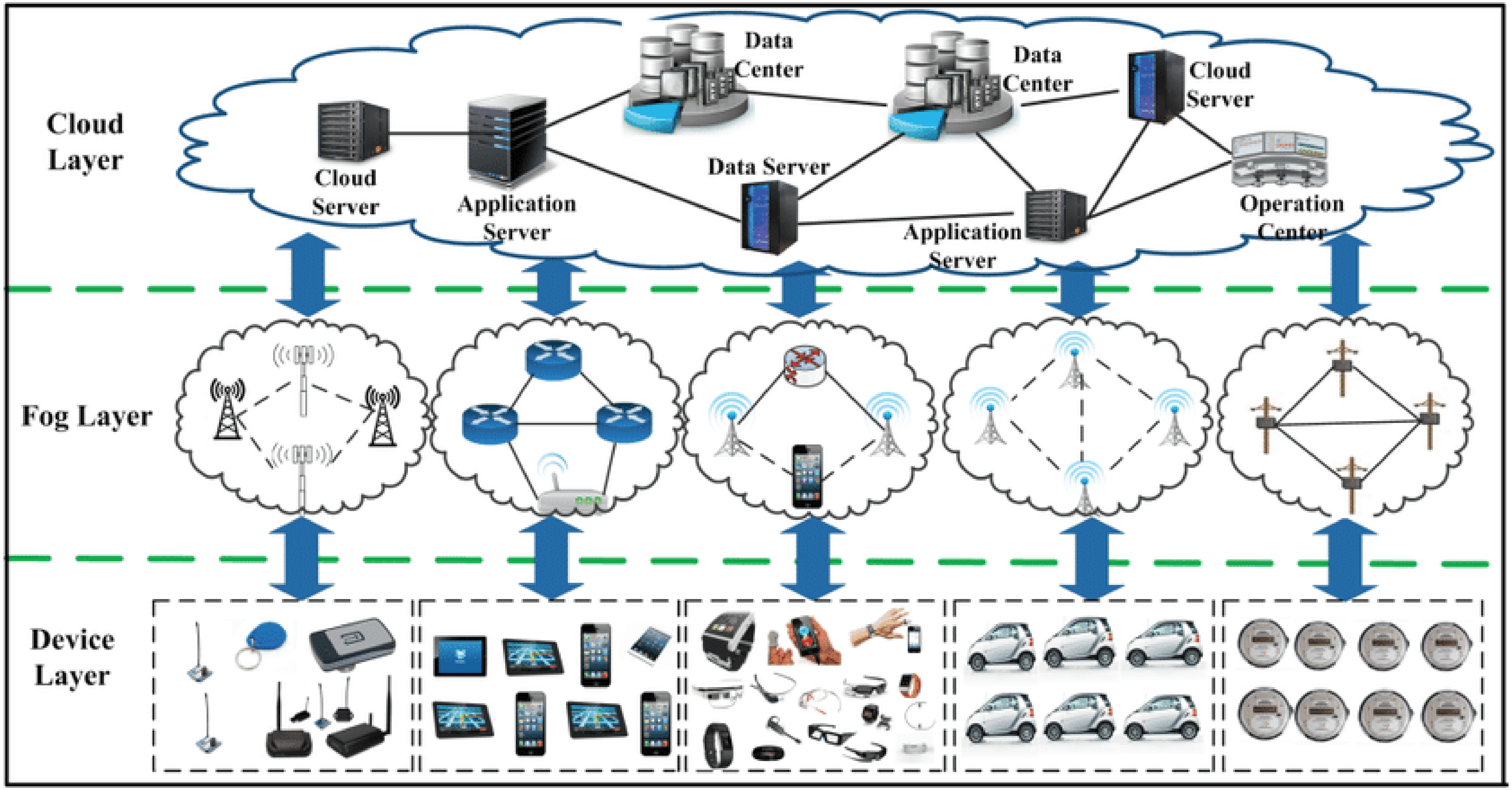
WINTER 2024 - 2025

Module 2 - Challenges in Federating Edge Resources

Introduction – Key issues in Cloud-to-Things



Federating Edge Resources - Introduction



Federating Edge Resources - Introduction

- The process of *connecting and sharing computing resources* located at the *edge of the network*.
- These resources can include *servers, storage, and networking equipment*.
- By federating these resources, organizations can create a *more distributed* and *scalable infrastructure* that can better support *applications* and services that require *low latency* and *high bandwidth*

Federating Edge Resources - Introduction

- Revisiting Edge of Network
  - The "edge of the network" refers to the ***boundary points*** where a ***computer network connects to other networks***.
  - It's essentially where one network ends and another begins.

Federating Edge Resources - Introduction

- Edge computing aims to ***improve application performance*** by ***decentralizing resources*** from cloud data centers to the network edge.
- Current edge deployments are often ***ad hoc, private,*** and sporadically ***distributed***, limiting their global impact.
- A ***federated edge***, connecting edge deployments across regions, is ***needed*** for ***global accessibility and computational fairness*** but faces ***technical, social, legal, and geopolitical challenges***.

Federating Edge Resources - Challenges

- Two key challenges for federated edge are ***networking and management***.
- The networking challenge involves creating a ***dynamic networking environment compatible*** with federated edge scenarios.
- ***Software-defined networking (SDN) is a potential solution*** for ***orchestrating edge resources***, but global coordination across SDN domains is required in a federated context.
- Balancing local edge deployments with the federated infrastructure is critical, requiring a potential rethinking of the edge networking model and further development of SDN's east-west interface.

Federating Edge Resources - Challenges

- Management challenges are crucial for seamless service delivery in large-scale edge computing infrastructures.
- Current edge deployments assume ***service cloning or availability*** on alternate nodes, which is insufficient for federated edge scenarios.
- Rapid service migration between nodes based on demand is needed but faces challenges due to ***high overhead and unsuitability*** for resource-constrained edge environments.
- Key management issues include benchmarking, provisioning, discovery, scaling, and migration.

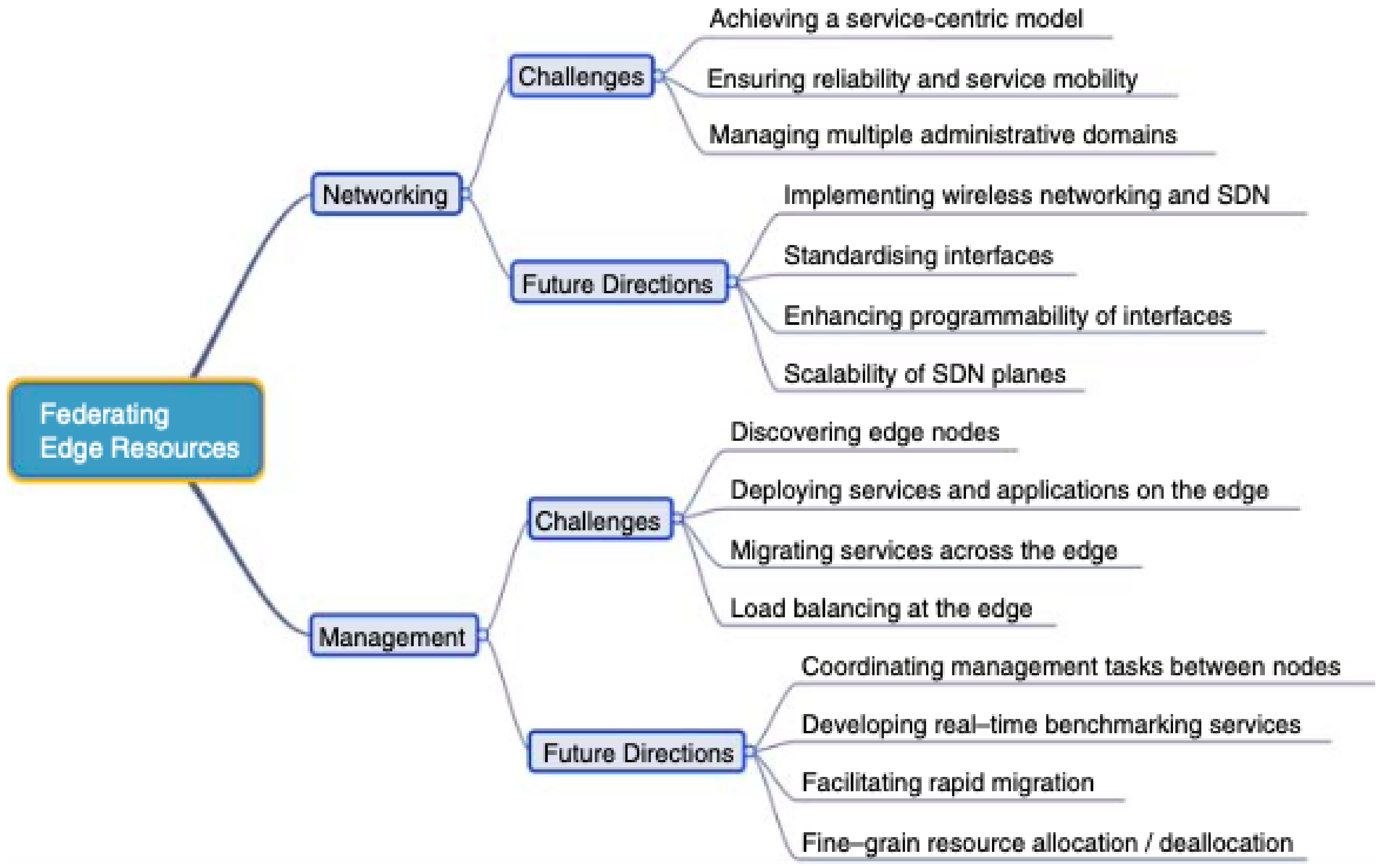
Federating Edge Resources - Challenges

- Resource challenges exist at both hardware and software levels.
- Hardware solutions are often application-specific, creating significant heterogeneity.
- Software solutions designed for cloud data centers have high overheads.

Federating Edge Resources - Challenges

- Modeling challenges arise from integrating the edge into the cloud ecosystem.
- Simulators are needed to investigate the implications of large-scale edge deployments, but current understanding of user-edge-cloud interactions is limited.

Federating Edge Resources - Challenges



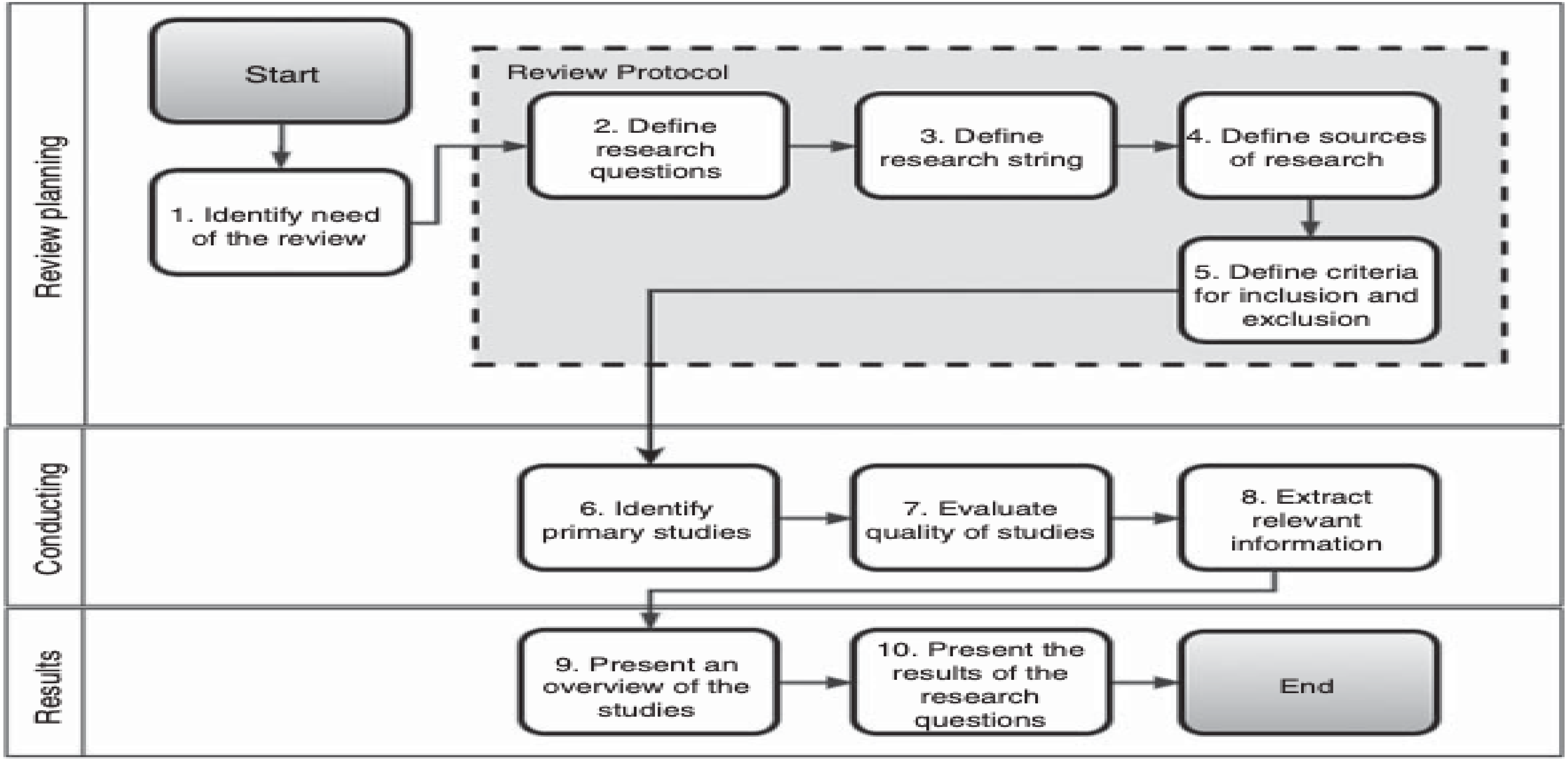
Federating Edge Resources - Challenges

Networking challenge	Why does it occur?	What is required?
User mobility	Keeping track of different mobility patterns	Mechanisms for application layer handover
QoS in a dynamic environment	Latency-intolerant services, dynamic state of the network	Reactive behavior of the network
Achieving a service-centric model	Enormous number of services with replications	Network mechanisms focusing on "what" instead of "where"
Ensuring reliability and service mobility	Devices and nodes joining the network (or leaving)	Frequent topology update, monitoring the servers and services
Managing multiple administrative domains	Heterogeneity, separate internal operations and characteristics, different service providers	Logically centralized, physically distributed control plane, vendor independency, global synchronization

Federating Edge Resources - Challenges

Management challenge	Why should it be addressed?	What is required?
Discovery of edge nodes	To select resources when they are geographically spread and loosely coupled	Lightweight protocols and handshaking
Deployment of service and applications	Provide isolation for multiple services and applications	Monitoring and benchmarking mechanisms in real-time
Migrating services	User mobility, workload balancing	Low overhead virtualization
Load balancing	To avoid heavy subscription on individual nodes	Auto-scaling mechanisms

Federating Edge Resources – Review Methodology



Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model

- Mathematical models that provide a *closed-form solution*, meaning the solution can be expressed as a *mathematical function*.

- Used to predict *resource requirements*, *assess workload behavior*, and *measure the impact of hardware/software* changes.

- Often rely on *approximations*, which must be carefully considered.

C2F2T – Analytical Model - Examples

Architecture Modeling
- Defines an architecture with vectors of associated features for physical and virtual components, using equations to calculate metrics like power consumption and time latency.

Gateway Modeling
- Considers the number of gateways, data received, and processing time, proposing equations to represent buffer occupancy and delay.

Scheduling Mechanism
- Proposes an analytical model for scheduling in IoT environments, accounting for variables like processor load, memory, and bandwidth.

QoS Prediction
- Presents a method to predict QoS of IoT services using a neighborhood collaborative filter and equations to predict latency, response time, and user network condition.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 1

- This model aims to ***manage*** sensor applications by a gateway, considering the resource requirements of these applications.
- Equation mathematically represents the ***optimization problem*** formulated in the research

$$min.\varphi = \sum_{i=1}^{n} \int_{0}^{T} r_i w_i(t)[1 - s_i(t)]dt$$

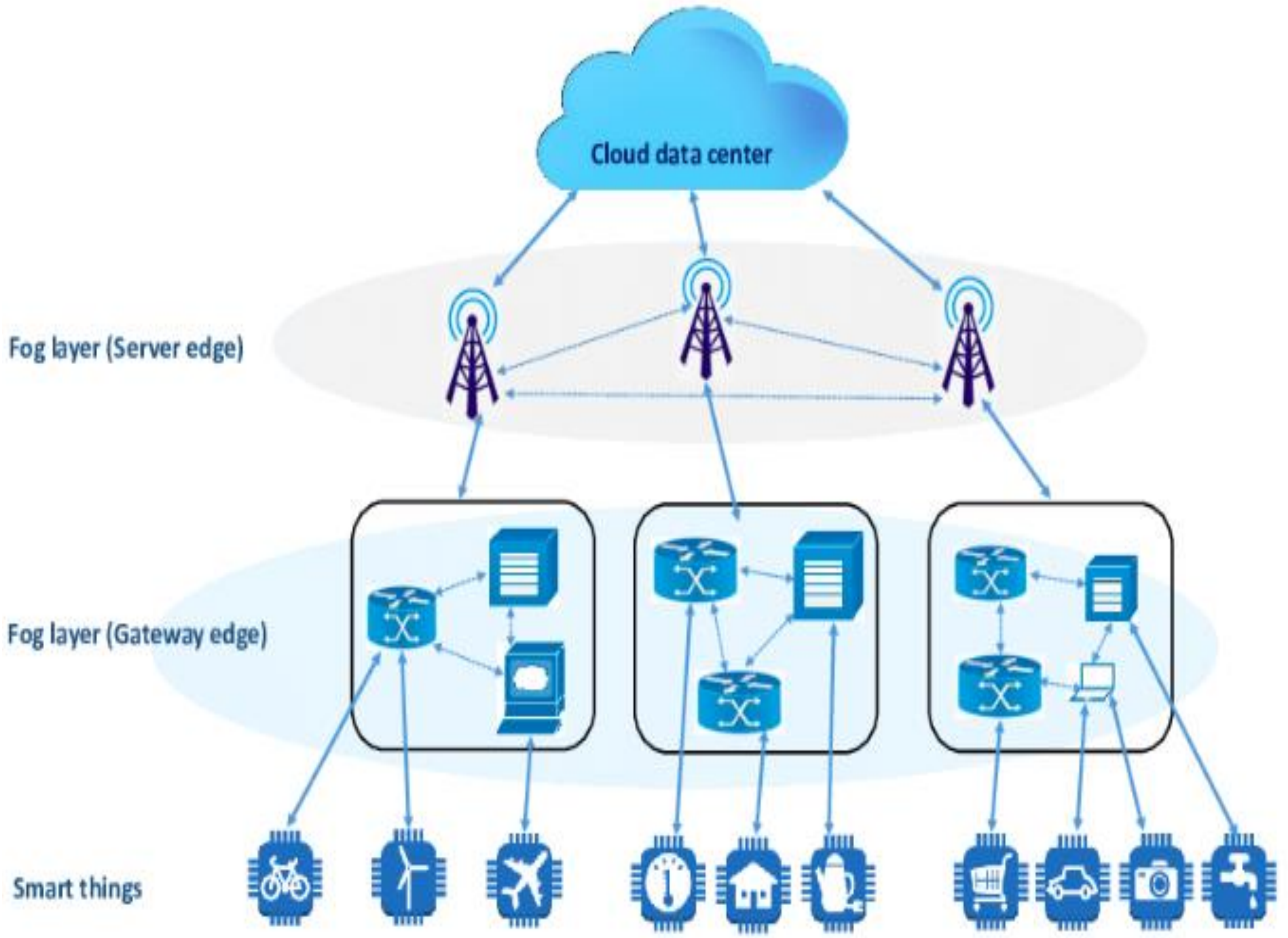
$$s.t.(1) \sum_{i=1}^{n} u_i s_i(t) \le \alpha_c u_c$$

$$(2) \sum_{i=1}^{n} c_i s_i(t) \le \alpha_m c_m$$

$$(3) \sum_{i=1}^{n} b_i^I s_i(t) \le \alpha_I b_I$$

$$(4) \sum_{i=1}^{n} b_i^O s_i(t) \le \alpha_O b_O$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 1



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$$(4) \sum_{i=1}^{n} b_i^O s_i(t) \le \alpha_O b_O$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 1

- A: Set of n applications
- a_i: Individual application, where i = 1, 2, ..., n
- R_i: Resource requirement vector for application a_i
- r_i: Priority of application a_i, ranging from 0 to 1
- $w_i(t)$: Required work status of application a_i at time t
- $s_i(t)$: Actual working status of application a_i at time t

$$min.\varphi = \sum_{i=1}^{n} \int_{0}^{T} r_i w_i(t)[1 - s_i(t)]dt$$

$$s.t.(1) \sum_{i=1}^{n} u_i s_i(t) \le \alpha_c u_c$$

$$(2) \sum_{i=1}^{n} c_i s_i(t) \le \alpha_m c_m$$

$$(3) \sum_{i=1}^{n} b_i^I s_i(t) \le \alpha_I b_I$$

$$(4) \sum_{i=1}^{n} b_i^O s_i(t) \le \alpha_O b_O$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 1

- ***n Applications:***
  - Represented by the set A = {a1(Ri, r_i, $w_i(t)$, $s_i(t)$), a2(R2, r_2, $w_2(t)$, $s_2(t)$), ..., an(Rn, r_n, $w_n(t)$, $s_n(t)$)}
- ***Each application (ai) is characterized by:***
  - R_i: Resource requirement vector (specifies the resources needed by the application)
  - $r_i \in [0, 1]$: Priority of the application (a value between 0 and 1)
  - $w_i(t)$: Required work status of the application at time 't'
  - $s_i(t)$: Actual working status of the application at time 't'

$$min.\varphi = \sum_{i=1}^{n} \int_{0}^{T} r_i w_i(t)[1 - s_i(t)]dt$$

$$s.t.(1) \sum_{i=1}^{n} u_i s_i(t) \le \alpha_c u_c$$

$$(2) \sum_{i=1}^{n} c_i s_i(t) \le \alpha_m c_m$$

$$(3) \sum_{i=1}^{n} b_i^l s_i(t) \le \alpha_l b_l$$

$$(4) \sum_{i=1}^{n} b_i^O s_i(t) \le \alpha_O b_O$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 1

- ***Objective Function (min.φ):***
  - The goal is to minimize the value of 'φ', which likely represents the ***total waiting time of all tasks*** (applications) being processed by the gateway.
  - The ***integral within the summation*** calculates the *waiting time for each application over the evaluation time 'T'*.

$$min.\varphi = \sum_{i=1}^{n} \int_{0}^{T} r_i w_i(t)[1 - s_i(t)]dt$$

$$s.t.(1) \sum_{i=1}^{n} u_i s_i(t) \le \alpha_c u_c$$

$$(2) \sum_{i=1}^{n} c_i s_i(t) \le \alpha_m c_m$$

$$(3) \sum_{i=1}^{n} b_i^I s_i(t) \le \alpha_I b_I$$

$$(4) \sum_{i=1}^{n} b_i^O s_i(t) \le \alpha_O b_O$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 1

- ***Constraints:***
  - The equations below "s.t." represent constraints that must be satisfied during the scheduling process.
  - **Constraint 1**: Limits the total CPU utilization (u_c) to a certain threshold (α_c).
  - **Constraint 2**: Limits the total memory consumption (c_m) to a certain threshold (α_m).
  - **Constraint 3**: Limits the input bandwidth usage (b_l) to a certain threshold (α_l).
  - **Constraint 4**: Limits the output bandwidth usage (b_o) to a certain threshold (α_o).
  - **α_c, α_m, α_l, α_o**: These are overloading factors that indicate the extent to which the system can tolerate resource usage exceeding the ideal limits while still maintaining smooth operation.

$$min.\varphi = \sum_{i=1}^{n} \int_0^T r_i w_i(t)[1 - s_i(t)]dt$$

$$s.t.(1) \sum_{i=1}^{n} u_i s_i(t) \leq \alpha_c u_c$$

$$(2) \sum_{i=1}^{n} c_i s_i(t) \leq \alpha_m c_m$$

$$(3) \sum_{i=1}^{n} b_i^I s_i(t) \leq \alpha_I b_I$$

$$(4) \sum_{i=1}^{n} b_i^O s_i(t) \leq \alpha_O b_O$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 1

- Equation formulates ***an optimization problem*** where the objective is to ***minimize the total waiting time of sensor applications*** while ***ensuring that the gateway's resources*** (CPU, memory, bandwidth) are ***not overloaded***.
- The model considers factors like application
  - ***priorities,***
  - ***resource requirements, and***
  - ***desired work status to schedule applications efficiently.***

$$min.\varphi = \sum_{i=1}^{n} \int_{0}^{T} r_i w_i(t)[1 - s_i(t)]dt$$

$$s.t.(1) \sum_{i=1}^{n} u_i s_i(t) \le \alpha_c u_c$$

$$(2) \sum_{i=1}^{n} c_i s_i(t) \le \alpha_m c_m$$

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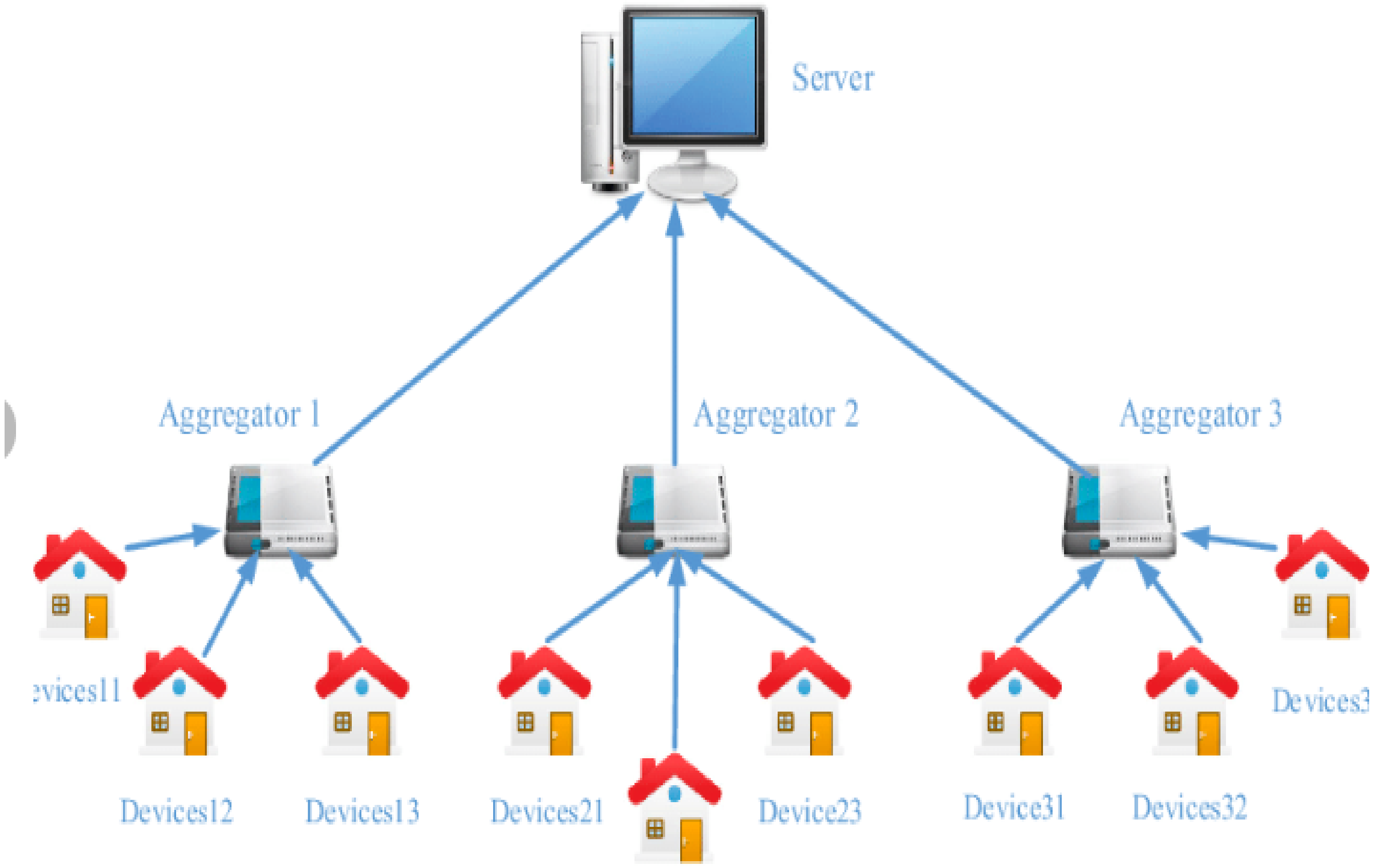
Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 2



Functions of Aggregator

- Data Collection
- Data Processing and Analysis
- Command and Control
- Data Storage and Management
- Communication Gateway

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 2

- Analytic expressions for representing the ***expected energy consumption of devices*** and ***a cloud billing*** method for a group of devices on an IoT aggregator.

$$E_{\exp} = E[\Psi_e]g_e + i_e \int_0^{c_e E[\Psi_e]} (c_e E[\Psi_e] - \omega_e) P_e(\omega_e) d\omega_e$$

- Computes the energy consumption of each device over the monitoring period, T

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 2

- ***expected energy consumption of devices*** *t*akes into account:
  - Query data volume of devices ($E[\Psi_e]$) - it's the average amount of data that devices send or receive in a given time period.
  - Consumption rate (lg_e, joule-per-bit) - It represents the amount of energy consumed by a device to process or transmit one bit of data.
  - "Idle" energy consumption by each device (i_e) - Idle mode refers to the state when a device is not actively processing or transmitting data but is still powered on.
  - Threshold for activating "idle" mode ($c_e E[\Psi_e]$) - This threshold determines when a device should switch to idle mode based on its expected query data volume.
  - Probability density function of Ψ_e ($P_e(\omega_e)$) - the probability distribution of the query data volume of devices, which can be used to model and analyze the behavior of the devices.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 2

- Cloud Billing Cost Model

$$B_{\exp} = E[\Psi_b]g_b + i_b \int_0^{c_b} (c_b - \omega_b)P(\omega_b)d\omega_b + p_b \int_{c_b}^{\infty} (\omega_b - c_b)P(\omega_b)d\omega_b$$

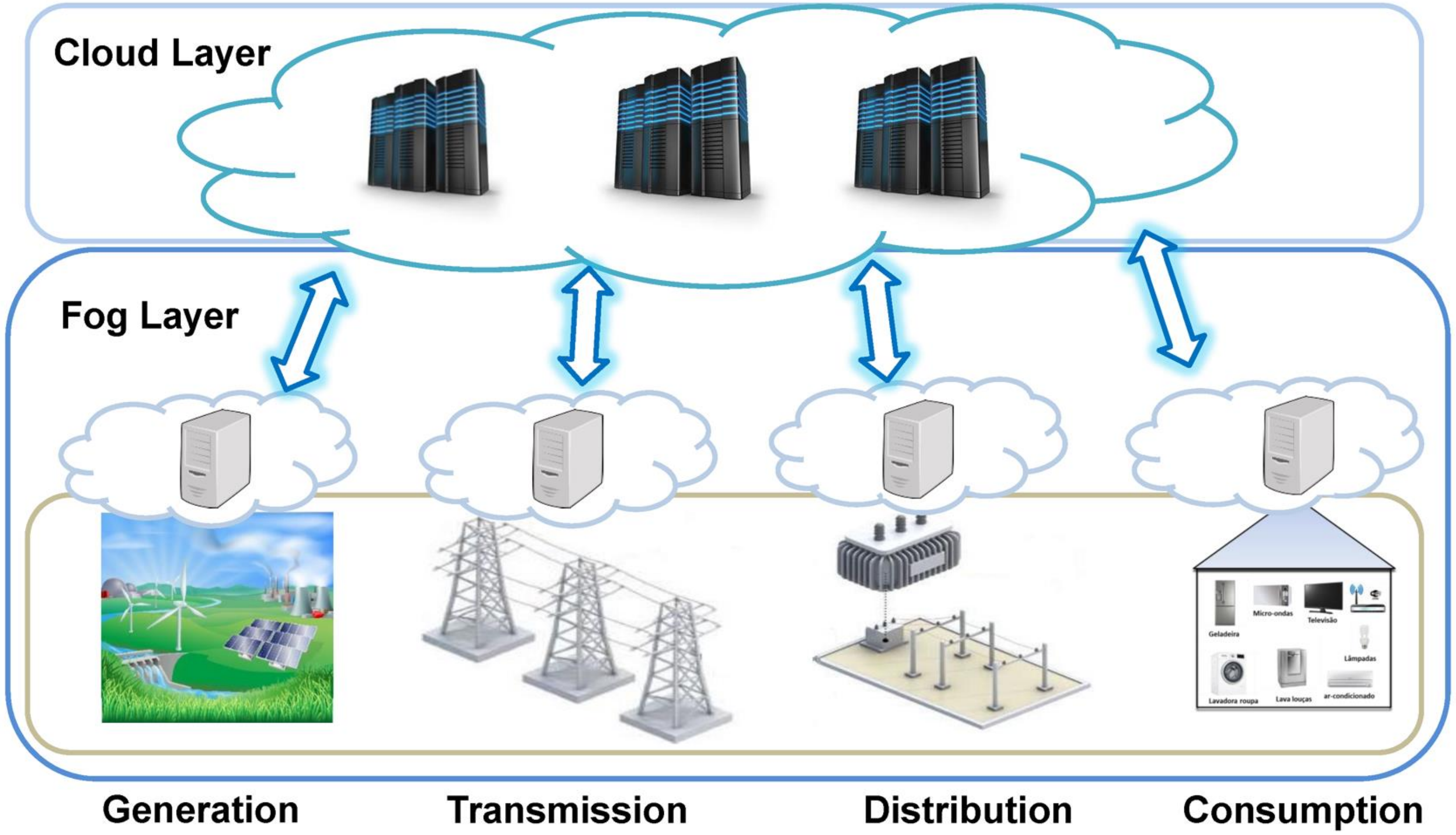
- Calculates the expected cloud billing cost for n aggregated query volumes from n devices

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 2

- Cloud Billing Cost Model takes into account
  - Transfer/storage costs ($E[\Psi_b]g_b$) - This represents the cost of transferring or storing data in the cloud.
  - Idle billing cost (first integral) - the cost incurred when devices are in idle mode but still connected to the cloud.
  - Active billing cost (second integral) - the cost incurred when devices are actively transmitting or receiving data
  - Cost of transferring one bit (i_b, dollars-per-query-bit) - the cost of transferring a single unit of data (one bit).
  - Coupling point defining the expected billing cost (c_b) - determines when the expected billing cost is incurred.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 3

- Proposes a strategy using ***fog computing with microgrids*** to ***reduce energy consumption*** of IoT applications.



Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 3

- Proposes a strategy using ***fog computing with microgrids*** to ***reduce energy consumption*** of IoT applications.
- Equation specifically ***calculates the energy consumed by an IoT service*** using cloud computing.

$$E_{IoT-cloud} = E_{GW-r} + E_{GW-t} + E_{net} + E_{DC}$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 3

- $E_{GW\text{-}r}$: Energy consumed by IoT gateways when receiving data from IoT devices and sensors.
- $E_{GW\text{-}t}$: Energy consumed by IoT gateways to transmit data to the cloud data center.
- E_{net}: Energy consumed by the transport network between IoT gateways and the cloud.
- E_{DC}: Energy consumed by components of the data center.

$$E_{IoT-cloud} = E_{GW-r} + E_{GW-t} + E_{net} + E_{DC}$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 3

- Equation calculates the ***energy consumption of communication between the IoT and the fog***.

$$E_{IoT-fog} = E_{GW-r} + E_{GW-c} + \beta(E_{GW-t} + E_{net} + E_{DC})$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 3

- E_{GW-r}: Energy consumed by IoT gateways when receiving data from IoT devices and sensors.
- E_{GW-t}: Energy consumed by IoT gateways to transmit data to the fog.
- E_{net}: Energy consumed by the transport network between IoT gateways and the fog.
- E_{DC}: Energy consumed by components of the fog.
- $\mathbf{E_{GW-c}}$: Energy consumed by IoT gateways for local computation and processing.
- $\boldsymbol{\beta}$: Ratio of the number of updates from the fog to the cloud for synchronization.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 4

- Proposes an analytical modelling approach to ***estimate the reliability of an IoT scenario*** in a smart home context.
- The algorithm estimates the reliability of the IoT service, which consists of n subsystems.
- **The calculation considers three factors:**
  - **Availability of programs (Ppr):** Availability of all k programs running on virtual machines.
  - **Availability of input files (Pf):** Availability of f input files for the programs.
  - **Reliability of each subsystem (ISR):** Reliability of each virtual machine being executed.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Analytical Model 4

$$R_s(t_b) = \prod_{i=1}^{N} ISR_i \; ISR_i \times \prod_{i=1}^{f} P_f(i) \times \prod_{i=1}^{k} P_{pr}(i)$$

Cloud-to-Fog-to-Internet of Things (C2F2T) – Other Analytical Model

Model Name	Purpose	Use case	Calculated
Fuzzy Ontology	Fault classification in WSNs; Handles uncertainty in sensor data; Improves fault diagnosis accuracy.	Fault types in WSNs, their relationships, and membership degrees.	Degree of membership of a sensor reading to different fault types; Most probable fault type.
Bayesian Probability	Predicting future events in IoT applications; Handling uncertainty in predictions; Incorporating prior knowledge and updating with new data.	Occurrence of future events in IoT systems (e.g., flight delays, equipment failures).	Probabilities of future events; Likelihood of chain events occurring.
Markov Chains	Modeling energy consumption in IoT devices; Predicting energy consumption patterns; Optimizing energy usage.	Device states (e.g., active, idle, sleep), state transitions, and energy consumption in each state.	Long-term energy consumption of devices; Optimal energy-saving strategies.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Other Analytical Model

Model Name	Purpose	Use case	Calculated
Integer Linear Programming (ILP) & Mixed-Integer Linear Programming (MILP)	Minimizing cost in IoT and cloud computing; Optimizing resource allocation, task scheduling, and network design.	Resource allocation, task scheduling, network design, and other optimization problems.	Optimal resource allocation, task schedules, network configurations to minimize cost.
Petri Nets	Modeling and analyzing the dynamics of C2F2T systems; Representing data flow, task execution, resource allocation, and synchronization; Verifying system correctness and performance.	Data flow, task execution, resource allocation, and synchronization in C2F2T systems.	System performance metrics (e.g., throughput, latency, resource utilization); Potential bottlenecks; System correctness and reliability.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Other Analytical Model

- Presents a model to evaluate the security level of C2T systems, focusing on the flow of information and system states.
- Uses a knapsack problem (MKP) to optimize service allocation in C2F2T scenarios, considering load balancing, delay, and energy consumption.
- Formulates an optimization problem to maximize cloud service utilities and obtain optimal payment and computation offloading.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Other Analytical Model

- Proposes analytical modeling to represent QoS of IoT composite services and optimize cost with QoS constraints.
- Uses analytical models to decide whether computing should be offloaded to IoT devices or the cloud, considering QoS and energy levels.
- Compares the performance of fog architecture against traditional cloud computing using analytical models.
- Present analytical models to represent mobile nodes connected to cloud computing, considering device movement and architecture components

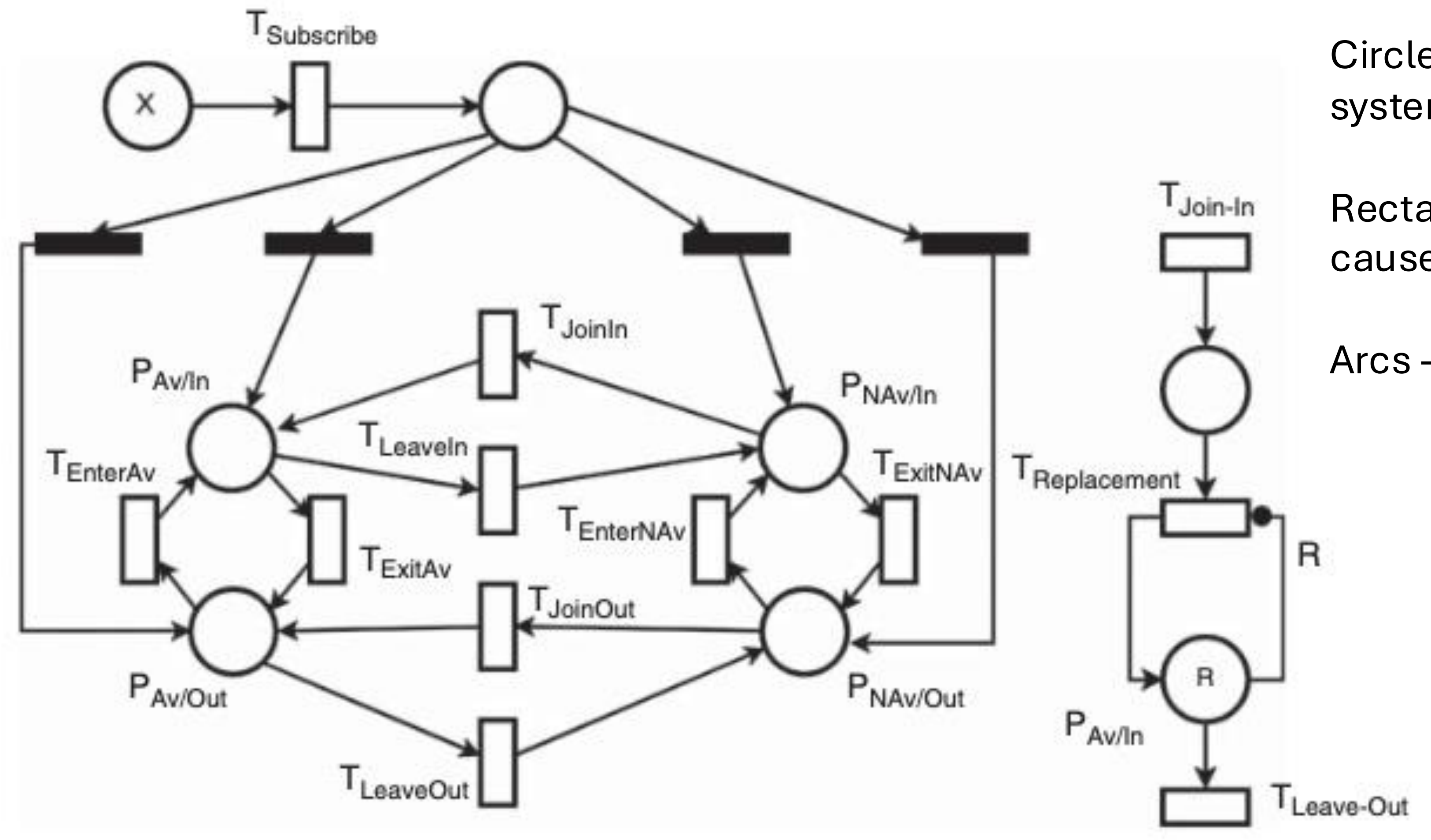
Cloud-to-Fog-to-Internet of Things (C2F2T) – Petri Net Models

- A powerful model for representing and analyzing systems (performance, dependability).
- Offer flexibility in modeling (Markovian and non-Markovian distributions).
- Can represent systems with fewer states compared to Markov chains.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Petri Net Models

- Petri nets are a powerful formalism for modeling concurrent systems
  - **Mobile Devices**: Each mobile device can be represented as a place in the Petri net. Tokens in the place indicate the availability of the device for sensing and data transmission.
  - **Sensing Tasks**: Sensing tasks, such as collecting location data, measuring temperature, or capturing images, can be modeled as transitions.
  - **Data Flow**: The movement of data from the mobile devices to the central server or processing unit can be represented by directed arcs connecting places and transitions.
  - **User Interactions**: User actions, such as initiating a sensing task, approving data sharing, or providing feedback, can be modeled as transitions.
  - **System States**: Different states of the MCS system, such as idle, active sensing, data transmission, and data processing, can be represented by different markings of the Petri net (i.e., the distribution of tokens in the places).

Cloud-to-Fog-to-Internet of Things (C2F2T) – Petri Net Models



(a) MCS. (b) MCSaaS.

Circle - the state or condition of the system

Rectangle - model events or actions that cause state changes.

Arcs - indicate the flow of tokens

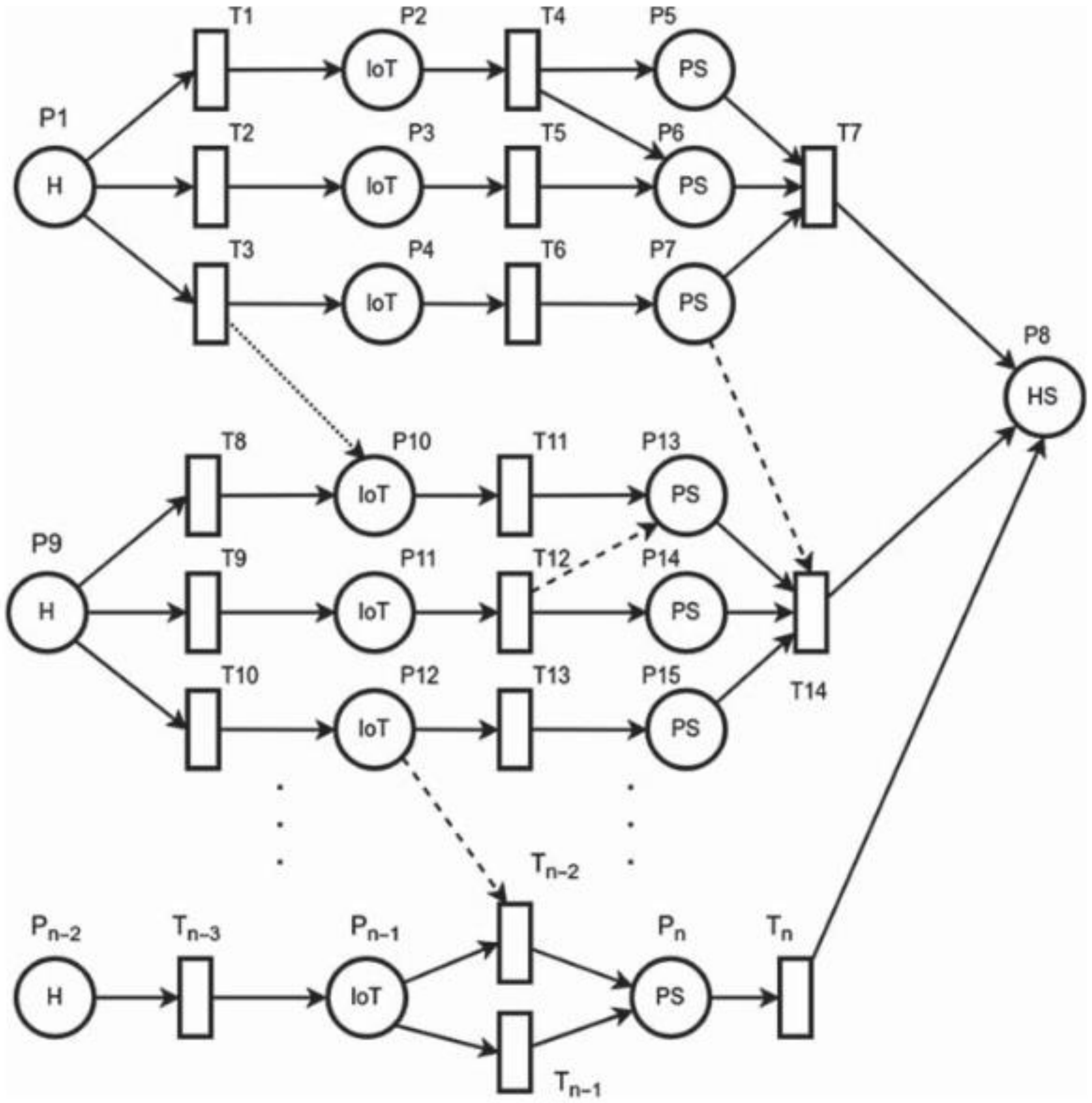
Cloud-to-Fog-to-Internet of Things (C2F2T) – Petri Net Models

- Investigated a ***mobile crowdsensing framework***.
- Used Petri Nets to compare the proposed framework with standard architectures.
- Modeled node states (availability, position) and transitions between them.
- Highlighted the complexity of the proposed framework due to added modules.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Petri Net Models

- Focused on ***data traceability in a C2T*** (Continuous Care and Treatment) scenario.
- Employed a Petri Net to map and match device data to users.
- Used the Petri Net to represent the behavior of a wearable IoT architecture.
- Modeled data sources and events (e.g., new medical data readings).

Cloud-to-Fog-to-Internet of Things (C2F2T) – Petri Net Models



- A method for solving optimization problems where all variables are integers.
- Objective function and constraints are linear.
- Mixed-Integer Linear Programming (MILP):
- An extension of ILP that allows both continuous and integer variables.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Integer Linear Programming

- Used Integer Linear Programming (ILP) to minimize the financial cost of a fog-cloud architecture in a MAN.
- Considered costs related to processing, storage, upstream/downstream traffic, and network capacity.
- Represented application profiles and node characteristics as vectors.

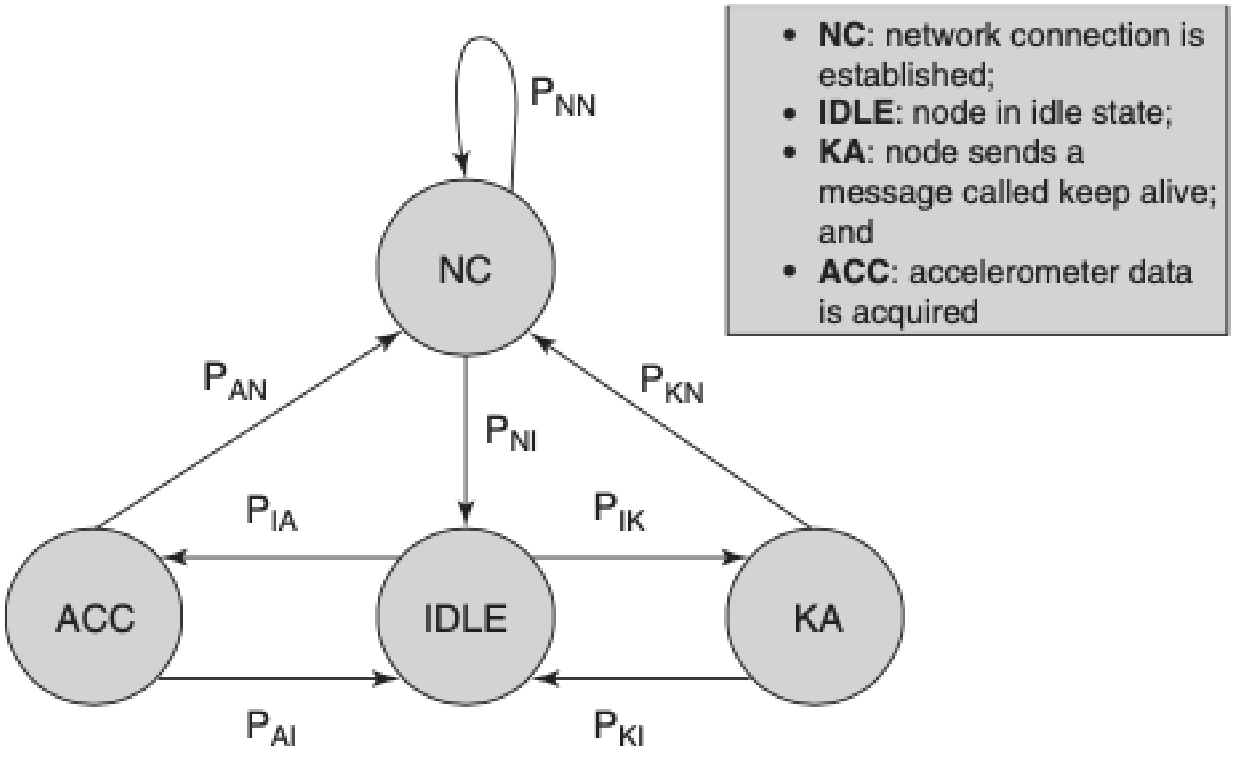
Cloud-to-Fog-to-Internet of Things (C2F2T) – Integer Linear Programming

- Employed MILP to minimize energy consumption in an IoT architecture with mini-clouds.
- Considered traffic-induced and processing-induced energy consumption.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Other Approaches

➢Markov Chains:

- ➢Model sequences of events where the current state depends only on the previous state.
- ➢Used in [21] to model energy consumption in an IoT-based collision detection system.
- ➢Energy consumption is estimated by calculating the probability of each system state.

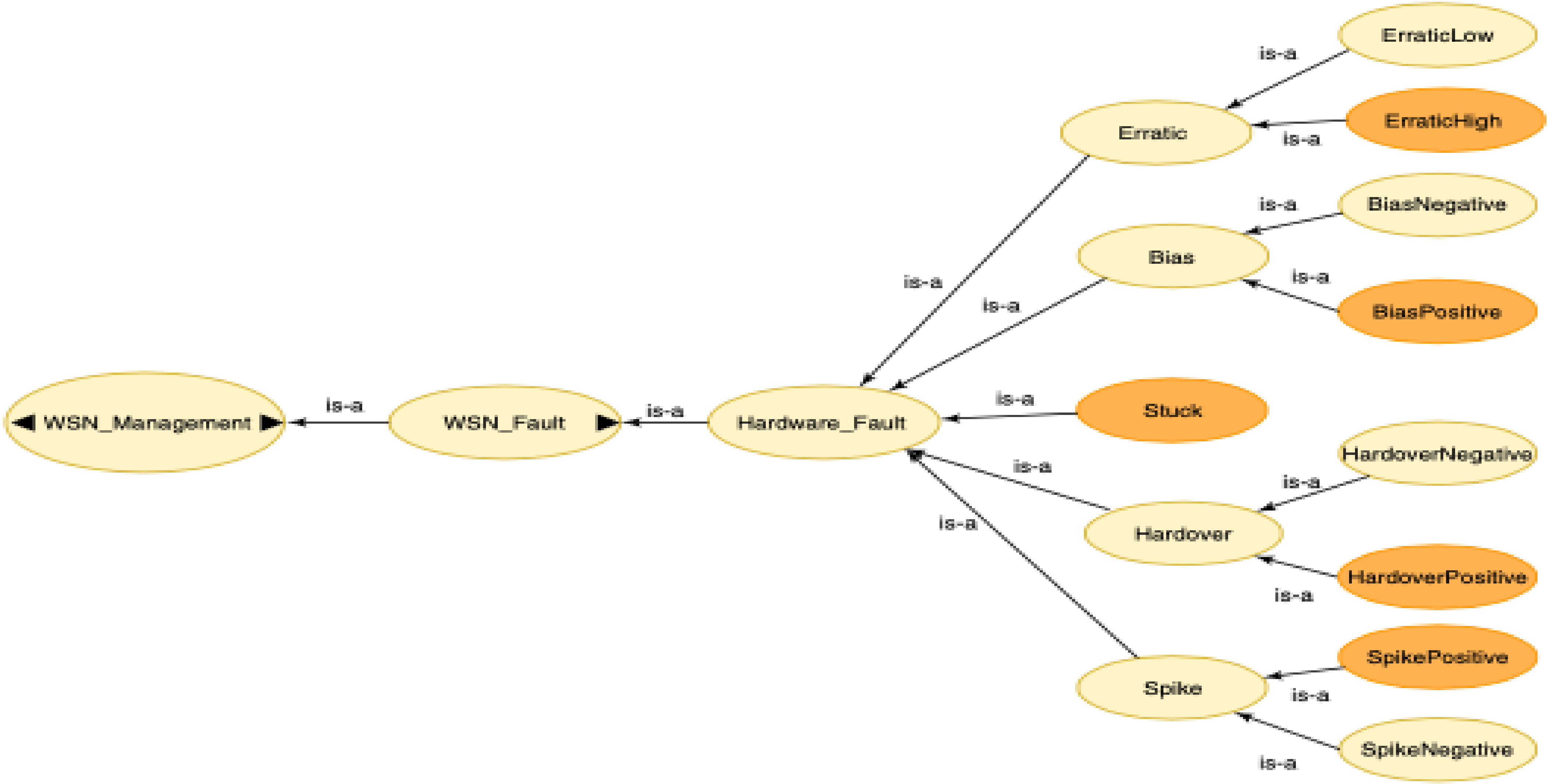


Cloud-to-Fog-to-Internet of Things (C2F2T) – Other Approaches

- **Bayesian Probability:**
  - Represents the probability of an event considering prior knowledge and available information.
  - Used in [24] to predict future events in IoT applications.
  - Focuses on calculating conditional probabilities of events.

- **Fuzzy-Ontology:**
  - Combines ontology (for describing system components and their relationships) with fuzzy logic (for handling uncertainty and imprecision).
  - Used in [17] to analyze faults in WSN scenarios.
  - Provides a flexible framework for detecting and categorizing faults from different perspectives.

Cloud-to-Fog-to-Internet of Things (C2F2T) – Other Approaches



Integrated C2F2T Literature by Modeling Technique

Modelling Technique	Use Case Scenario	Application	Metrics	Key Findings/Objectives
Analytical Model	Resource Management	Energy Consumption of Devices in Cloud-IoT Context	Energy Consumption	Analyze the relationship between energy consumption of IoT devices and cloud billing costs.
Analytical Model	Resource Management	Energy Efficiency of IoT Devices	Energy Consumption	Improve energy efficiency of IoT devices.
Analytical Model	Resource Management	Energy Consumption of WSN	Energy Consumption	Analyze energy consumption in a WSN.
Analytical Model	Resource Management	Energy Consumption in Fog-IoT Integration	Energy Consumption	Reduce energy consumption by integrating IoT and fog computing.
Analytical Model	Resource Management	Resource Allocation in C2F2T	Energy Consumption, Resource Usage	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.
Analytical Model	Resource Management	QoS in IoT and Cloud-IoT	Response Time, Reliability, Availability	Calculate QoS of IoT services and predict QoS for IoT and cloud services.

Integrated C2F2T Literature by Modeling Technique

Modelling Technique	Use Case Scenario	Application	Metrics	Key Findings/Objectives
Analytical Model	Smart Cities	Air Quality Sensing with Mobile Devices	Energy Consumption, Percentage Energy Saved	Model energy consumption in a mobile crowdsensing application.
Analytical Model	Smart Cities	Vehicle-to-Cloud Monitoring	QoS (Object Detection Accuracy), Energy Consumption	Determine optimal processing location (edge or cloud) for video data.
Analytical Model	Smart Cities	Smart Living Spaces with Cloud Access	Scheduling Performance (Scalability, Waiting Time, Throughput)	Develop a scheduling model for managing sensor applications in a smart living space.
Analytical Model	Smart Cities	Fire Alarm System	Reliability	Evaluate the reliability of a fire alarm system in an IoT context.
Analytical Model	Wireless Sensor Networks (WSN)	Fault Tolerance	Packet Delivery Latency	Model a WSN for sensing as a service integrated with IoT and cloud.
Analytical Model	Health	Health Monitoring	-	Develop a framework for sharing IoT devices in health monitoring.

Integrated C2F2T Literature by Modeling Technique

Modelling Technique	Use Case Scenario	Application	Metrics	Key Findings/Objectives
Analytical Model	Health	Medical Research	Security (Opacity)	Analyze security of information flow in cloud-integrated IoT systems.
Analytical Model	Generic IoT/Fog/Cloud	Three-tier C2F2T Model	Energy Consumption, Latency	Represent a generic C2F2T architecture and analyze energy consumption and latency.
Analytical Model	Generic IoT/Fog/Cloud	Fog Computing for Improved Response Time	Effective Arrival Rate, Average System Response, Average Buffer Occupancy	Improve response time of IoT systems using fog computing.
Analytical Model	Generic IoT/Fog/Cloud	Generic Fog Architecture	Energy Consumption, Latency	Compare performance of fog architecture against traditional cloud computing.
Analytical Model	Generic IoT/Fog/Cloud	Edge Offloading	Resource Consumption (Cloud and IoT Device Consumption)	Improve mobile user experience by offloading computation to the edge.
Petri Net	Smart Cities	Crowdsensing in Smart Traffic	System Effectiveness	Evaluate a mobile crowdsensing-as-a-service architecture.

Integrated C2F2T Literature by Modeling Technique

Modelling Technique	Use Case Scenario	Application	Metrics	Key Findings/Objectives
Petri Net	Health	Wearable IoT for Healthcare	Reliability	Manage traceability of data in a wearable IoT architecture.
Integer Linear Programming (ILP)	Resource Management	Resource Allocation in C2F2T	Energy Consumption, Resource Usage	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.
Other Approaches	Smart Cities	Airplane Monitoring	-	Predict flight delays based on equipment data.
Other Approaches	Wireless Sensor Networks (WSN)	Fault Tolerance	-	Verify fault tolerance in large-scale WSN using service-oriented applications.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Use case Scenarios

- **Health:**
  - Security of information flow in medical applications
  - Sharing IoT devices for health monitoring
  - Wearable IoT architecture for healthcare
- WSN:
  - Fault tolerance in large-scale WSNs
  - WSN model for sensing as a service

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Use case Scenarios

- **Smart Cities:**
  - Mobile crowdsensing for smart traffic
  - Air quality monitoring using mobile sensors
  - Vehicle-to-cloud monitoring with edge computing
  - Fire alarm system
  - Flight delay prediction
  - Smart living space scheduling
- **Resource Management:**
  - Energy consumption of IoT devices
  - Resource allocation in C2F2T context
  - QoS in IoT and cloud-IoT environments

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Use case Scenarios

- Other/Generic:
  - General C2F2T architecture
  - Improving response time in IoT systems
  - Generic fog architecture
  - Offloading computing to the edge

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Use case Scenarios

Scenarios	Applications
Generic	Generic IoT/fog/cloud applications Edge offloading computing
Health	Health monitoring Medical research
WSN	Sensing as a service Fault tolerance
Smart cities	Mobile crowd sensing as a service Smart traffic Smart living Airplane monitoring
Resource management	Resource allocation Energy efficiency Quality of service (QoS)

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Use case Scenarios

Use Case Scenario	Application	Modelling Technique	Metrics	Key Findings/Objectives
Resource Management	Energy Consumption of Devices in Cloud-IoT Context	Analytical Model	Energy Consumption	Analyze the relationship between energy consumption of IoT devices and cloud billing costs.
Resource Management	Energy Efficiency of IoT Devices	Analytical Model	Energy Consumption	Improve energy efficiency of IoT devices.
Resource Management	Energy Consumption of WSN	Analytical Model	Energy Consumption	Analyze energy consumption in a WSN.
Resource Management	Energy Consumption in Fog-IoT Integration	Analytical Model	Energy Consumption	Reduce energy consumption by integrating IoT and fog computing.
Resource Management	Resource Allocation in C2F2T	Analytical Model, Integer Linear Programming	Energy Consumption, Resource Usage	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.
Resource Management	QoS in IoT and Cloud-IoT	Analytical Model	Response Time, Reliability, Availability	Calculate QoS of IoT services and predict QoS for IoT and cloud services.
Smart Cities	Crowdsensing in Smart Traffic	Petri Net	System Effectiveness	Evaluate a mobile crowdsensing-as-a-service architecture.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Use case Scenarios

Use Case Scenario	Application	Modelling Technique	Metrics	Key Findings/Objectives
Smart Cities	Air Quality Sensing with Mobile Devices	Analytical Model	Energy Consumption, Percentage Energy Saved	Model energy consumption in a mobile crowdsensing application.
Smart Cities	Vehicle-to-Cloud Monitoring	Analytical Model	QoS (Object Detection Accuracy), Energy Consumption	Determine optimal processing location (edge or cloud) for video data.
Smart Cities	Smart Living Spaces with Cloud Access	Analytical Model	Scheduling Performance (Scalability, Waiting Time, Throughput)	Develop a scheduling model for managing sensor applications in a smart living space.
Smart Cities	Fire Alarm System	Analytical Model	Reliability	Evaluate the reliability of a fire alarm system in an IoT context.
Smart Cities	Airplane Monitoring	Bayesian Model	-	Predict flight delays based on equipment data.
Wireless Sensor Networks (WSN)	Sensing as a Service	Fault Fuzzy-Ontology	-	Verify fault tolerance in large-scale WSN.
Wireless Sensor Networks (WSN)	Fault Tolerance	Analytical Model	Packet Delivery Latency	Model a WSN for sensing as a service integrated with IoT and cloud.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Use case Scenarios

Use Case Scenario	Application	Modelling Technique	Metrics	Key Findings/Objectives
Health	Health Monitoring	Analytical Model	-	Develop a framework for sharing IoT devices in health monitoring.
Health	Medical Research	Analytical Model	Security (Opacity)	Analyze security of information flow in cloud-integrated IoT systems.
Health	Wearable IoT for Healthcare	Petri Net	Reliability	Manage traceability of data in a wearable IoT architecture.
Generic IoT/Fog/Cloud	Three-tier C2F2T Model	Analytical Model	Energy Consumption, Latency	Represent a generic C2F2T architecture and analyze energy consumption and latency.
Generic IoT/Fog/Cloud	Fog Computing for Improved Response Time	Analytical Model	Effective Arrival Rate, Average System Response, Average Buffer Occupancy	Improve response time of IoT systems using fog computing.
Generic IoT/Fog/Cloud	Generic Fog Architecture	Analytical Model	Energy Consumption, Latency	Compare performance of fog architecture against traditional cloud computing.
Generic IoT/Fog/Cloud	Edge Offloading	Analytical Model	Resource Consumption (Cloud and IoT Device Consumption)	Improve mobile user experience by offloading computation to the edge.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics

Metric	Variations
Energy consumption	Devices power consumption Energy efficiency Percentage energy saved
Performance	Latency Effective arrival rate Average system response System effectiveness Cache performance
Resource consumption	Usage of devices Average buffer occupancy
Costs	Itemized costs Operational costs
QoS	Quality of pattern recognition in images Response time Reliability
Security	Opacity

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics

Metric	Use Case Scenario	Application	Modeling Technique	Key Findings/Objectives
Energy Consumption	Resource Management	Energy Consumption of Devices in Cloud-IoT Context	Analytical Model	Analyze the relationship between energy consumption of IoT devices and cloud billing costs.
Energy Consumption	Resource Management	Energy Efficiency of IoT Devices	Analytical Model	Improve energy efficiency of IoT devices.
Energy Consumption	Resource Management	Energy Consumption of WSN	Analytical Model	Analyze energy consumption in a WSN.
Energy Consumption	Resource Management	Energy Consumption in Fog-IoT Integration	Analytical Model	Reduce energy consumption by integrating IoT and fog computing.
Energy Consumption	Resource Management	Resource Allocation in C2F2T	Analytical Model, Integer Linear Programming	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.
Energy Consumption	Smart Cities	Air Quality Sensing with Mobile Devices	Analytical Model	Model energy consumption in a mobile crowdsensing application.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics

Metric	Use Case Scenario	Application	Modeling Technique	Key Findings/Objectives
Energy Consumption	Smart Cities	Vehicle-to-Cloud Monitoring	Analytical Model	Determine optimal processing location (edge or cloud) for video data.
Energy Consumption	Generic IoT/Fog/Cloud	Three-tier C2F2T Model	Analytical Model	Represent a generic C2F2T architecture and analyze energy consumption.
Energy Consumption	Generic IoT/Fog/Cloud	Generic Fog Architecture	Analytical Model	Compare performance of fog architecture against traditional cloud computing.
Performance	Resource Management	QoS in IoT and Cloud-IoT	Analytical Model	Calculate QoS of IoT services and predict QoS for IoT and cloud services.
Performance	Smart Cities	Crowdsensing in Smart Traffic	Petri Net	Evaluate a mobile crowdsensing-as-a-service architecture.
Performance	Smart Cities	Smart Living Spaces with Cloud Access	Analytical Model	Develop a scheduling model for managing sensor applications in a smart living space.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics

Metric	Use Case Scenario	Application	Modeling Technique	Key Findings/Objectives
Performance	**Generic IoT/Fog/Cloud**	**Fog Computing for Improved Response Time**	**Analytical Model**	**Improve response time of IoT systems using fog computing.**
Resource Consumption	Resource Management	Resource Allocation in C2F2T	Analytical Model, Integer Linear Programming	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.
Resource Consumption	Generic IoT/Fog/Cloud	Edge Offloading	Analytical Model	Improve mobile user experience by offloading computation to the edge.
Cost	Resource Management	Energy Consumption of Devices in Cloud-IoT Context	Analytical Model	Analyze the relationship between energy consumption of IoT devices and cloud billing costs.
QoS	Resource Management	QoS in IoT and Cloud-IoT	Analytical Model	Calculate QoS of IoT services and predict QoS for IoT and cloud services.
QoS	Smart Cities	Vehicle-to-Cloud Monitoring	Analytical Model	Determine optimal processing location (edge or cloud) for video data.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics

Metric	Use Case Scenario	Application	Modeling Technique	Key Findings/Objectives
QoS	Smart Cities	Fire Alarm System	Analytical Model	Evaluate the reliability of a fire alarm system in an IoT context.
QoS	Health	Wearable IoT for Healthcare	Petri Net	Manage traceability of data in a wearable IoT architecture.
Security	Health	Medical Research	Analytical Model	Analyse security of information flow in cloud-integrated IoT systems.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics - **Energy consumption**

- **Energy consumption** is a major concern in C2F2T systems.
- Factors influencing energy consumption:
  - Device modes: Active, idle, transitions between states
  - Communication: Data transmission, reception, listening, sensing
  - Distance and medium: Communication distance and environment
  - Traffic: Data traffic between layers
  - Processing: Processing-induced energy consumption by VMs
  - Device movement: Mobility of devices
  - Synchronization: Synchronization between IoT devices, fog, and cloud
  - Data flow: Application data flow patterns
  - Power sources: Renewable energy sources like solar power

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics - **Performance**

- Latency:
  - **Processing latency**: Time taken for application tasks
  - **Transmission latency**: Communication delay for data transfer.
  - **Packet delivery latency**: Influenced by factors like hops, sleep intervals, and transmission time.
  - **Request processing delay**: Influenced by device computational complexity and application flow size.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics - **Performance**

➢Other Performance Metrics:

- ➢**Scalability**: Ability to handle increasing numbers of devices and data
- ➢**Waiting time**: Average time for applications to be processed
- ➢**Throughput:** Number of applications processed successfully within a given time
- ➢**Response time**: Time elapsed from data generation to processing by a gateway
- ➢**Processing efficiency**: Considers processing time, buffer occupancy, and gateway response times

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics - **Performance**

- **Evaluation Methods**
  - **Analytical models**: Mathematical models to analyze latency, delay, and processing efficiency
  - **Simulation**: Simulating system behavior to evaluate performance under different conditions
  - **Petri Nets**: Modeling system behavior and evaluating performance through state transitions

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics – **Resource Consumption**

- Careful consideration of where workloads should be processed.
- Focuses on optimizing service allocation in C2F2T infrastructures. It considers device resources and evaluates the advantages of fog computing.
- Assesses resource consumption in C2T scenarios. This includes both cloud infrastructure and IoT device consumption.
- IoT devices have limited computing and storage capacity.
- Evaluates the effectiveness of gateways by examining buffer occupancy. It models data arrival at gateways as a queuing system.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics - Cost

- Cloud-heavy applications significantly increase costs for both providers and consumers.
- Cost studies often overlap with energy consumption studies.
- Investigated billing costs associated with reserved computational resources for IoT queries. Billing costs are directly linked to the volume of queries generated by IoT devices.
- To minimize the operational cost of routing IoT traffic to the cloud. This includes costs associated with processing, storage, upstream/downstream traffic, local/global traffic, and link capacity.

- QoS is significantly impacted by various factors:
  - Network delays
  - Available computational resources
  - Number of IoT devices consuming resources
- Evaluating QoS for composite services:
  - Breaking down composite services into smaller, more granular services for individual QoS evaluation.

- QoS in video streaming applications:
  - Analyzes video streams from vehicles, defining QoS as the accuracy of object detection in video images.
  - Varying video resolution impacts CPU, memory, and bandwidth consumption.
  - Accuracy is defined as the probability of a detection result exceeding a certain threshold among multiple results.

- Reliability as a key aspect of QoS:
  - Emphasizes the importance of factors like service availability, input file availability, and reliability of individual subsystems for overall IoT system reliability.
  - Defines reliability in the context of security, specifically the ability of the traceability model to detect and prevent attacks.

Cloud-to-Fog-to-Internet of Things (C2F2T) – By Metrics – Quality of Service

- Security is addressed in only one of the mentioned articles ([15]).
- focuses on "opacity" in data flow within the C2T context.
  - Opacity is a uniform approach for defining security properties as predicates.
- Analytical models are used to verify opacity in a medical research application scenario.